3.3.4 Mass transport

Over large distances, efficient movement of substance to and from exchange surfaces is provided by mass transport.

3.3.4.1 Mass transport in animals

Content	Opportunities for skills development
The haemoglobins are a group of chemically similar molecules found in many different organisms. Haemoglobin is a protein with a quaternary structure. The role of haemoglobin and red blood cells in the transport of oxygen. The loading, transport and unloading of oxygen in relation to the oxyhaemoglobin dissociation curve. The cooperative nature of oxygen binding to show that the change in shape of haemoglobin caused by binding of the first oxygens makes the binding of further oxygens easier. The effects of carbon dioxide concentration on the dissociation of oxyhaemoglobin (the Bohr effect). Many animals are adapted to their environment by possessing different types of haemoglobin with different oxygen transport properties. The general pattern of blood circulation in a mammal. Names are required only of the coronary arteries and of the blood vessels entering and leaving the heart, lungs and kidneys. The gross structure of the human heart. Pressure and volume changes and associated valve movements during the cardiac cycle that maintain a unidirectional flow of blood. The structure of arteries, arterioles and veins in relation to their function. The structure of capillaries and the importance of capillary beds as exchange surfaces. The formation of tissue fluid and its return to the circulatory system. **Students should be able to:** • analyse and interpret data relating to pressure and volume changes during the cardiac cycle • analyse and interpret data associated with specific risk factors and the incidence of cardiovascular disease • evaluate conflicting evidence associated with risk factors affecting cardiovascular disease • recognise correlations and causal relationships.	**AT h** Students could design and carry out an investigation into the effect of a named variable on human pulse rate or on the heart rate of an invertebrate, such as *Daphnia*. **MS 2.2** Students could be given values of cardiac output (CO) and one other measure, requiring them to change the subject of the equation: $CO = \text{stroke volume} \times \text{heart rate}$
Required practical 5: Dissection of animal or plant gas exchange system or mass transport system or of organ within such a system.	**AT j**

3.3.4.2 Mass transport in plants

Content	Opportunities for skills development
Xylem as the tissue that transports water in the stem and leaves of plants. The cohesion-tension theory of water transport in the xylem. Phloem as the tissue that transports organic substances in plants. The mass flow hypothesis for the mechanism of translocation in plants. The use of tracers and ringing experiments to investigate transport in plants. **Students should be able to:** • recognise correlations and causal relationships • interpret evidence from tracer and ringing experiments and to evaluate the evidence for and against the mass flow hypothesis.	**AT b** Students could set up and use a potometer to investigate the effect of a named environmental variable on the rate of transpiration.